

2

Collection of Data

Collection of Data

Data is a collection of facts, figures, objects, symbols, and events from different sources.

Governments and organizations collect data using various methods to make better decisions.

Collection of Data

Without data, it would be difficult for organizations to make appropriate decisions, so data is collected from different audiences at various times.

Collection of Data

Definition:- Collection of data is gathering all the necessary information for the purpose of inquiry.

Whether you are performing research for business, governmental or academic purposes, data collection allows you to gain first-hand knowledge and original insights into your research problem.

Collection of Data (Contd. . . .)

Example:- a) An organization must collect data on product demand, customer preferences, and about competitors before launching a new product. If data is not collected beforehand, the organization's newly launched product may fail for many reasons, such as less demand and inability to meet customer needs.

Collection of Data (Contd. . . .)

Example:- b) in science the data resulting from experiment must be collected and analyzed.

Collection of Data (Contd. . . .)

- c) in education educators collect data and analyze to identify learning gaps, assess student progress, evaluate their own instruction and so on.
- d) a city government conducts a census to collect demographic data for planning infrastructure and social programs.

Collection of Data (Contd. . . .)

While methods and aims may differ between fields, the overall process of data collection remains largely the same. Before you begin collecting data, you need to consider:

- The **aim** of the research
- The **type of data** that you will collect
- The **methods and procedures** you will use to collect, store, and process the data

SCOPES OF STATISTICAL INVESTIGATIONS

SCOPES OF STATISTICAL INVESTIGATIONS

Definition 1:- A *population* is the set of all individuals of interest in a particular study.

Definition 2 :- A *census* is a survey in which observations are made on the entire population.

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

- That is if every element in the population can be listed, or enumerated, and observed, then a census is compiled.
- A census is “100%” survey

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

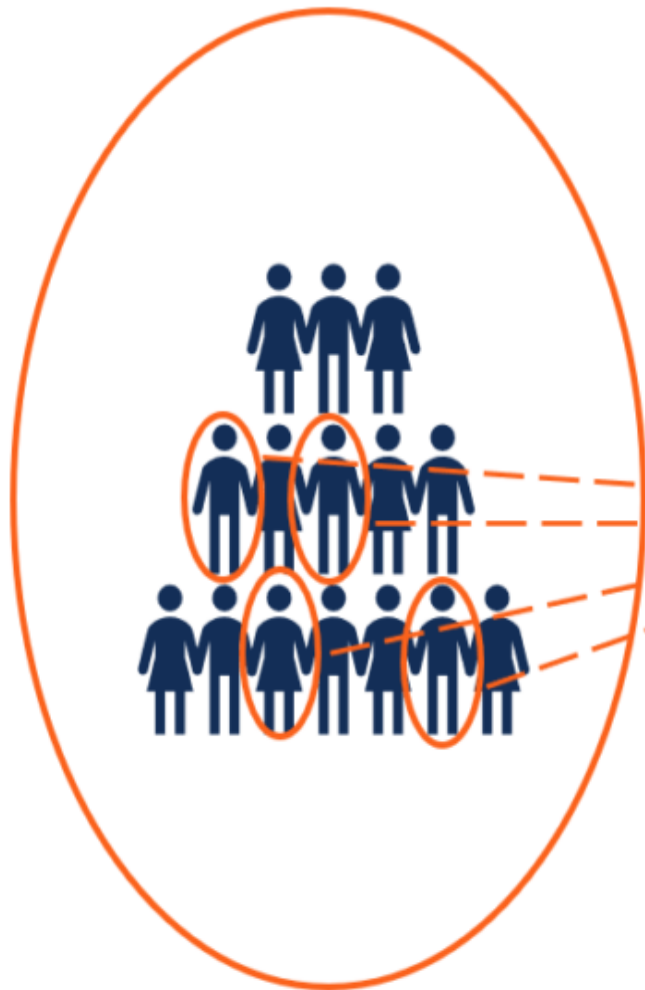
Definition 3 :- A *sample* is a set of individuals selected from a population, usually intended *to represent the population* in a research study.

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

Definition 4 :- *Sampling error* is the discrepancy, or amount of error, that exist between the sample statistic and the population parameter.

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

Population



Randomly
selected or biased
sample



Sampling
error
occurs

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

Categories of Sampling Errors

- *Population Specification Error* – Happens when the analysts do not understand who to survey. For example, for a survey of breakfast cereals, the population can be the mother, children, or the entire family.

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

- ***Selection Error*** – Occurs when the respondents' survey participation is self-selected, implying only those who are interested to respond. Selection errors can be reduced by encouraging participation.
- ***Sample Frame Error*** – Occurs when a sample is selected from the wrong population data.

SCOPES OF STATISTICAL INVESTIGATIONS (Contd. . . .)

- *Non-Response Error* – Occurs when a useful response is not obtained from the surveys. It may happen due to the inability to contact potential respondents or their refusal to respond.

ADVANTAGES AND DISADVANTAGES OF CENSUS AND SAMPLE SURVEY

Advantage and Disadvantage of census:

Advantage of census:

- More accurate than sample survey.
- Avoids sampling errors:

Disadvantage of census:

- ❑ Very expensive
- ❑ Time taking

Advantage and Disadvantage of census:

- ❑ Sometimes impossible due to different reasons.
- ❑ There is usually a significant delay between when the data are collected and the results released. Typically this delay is between 18 months to two years, and means that the census only offers a snapshot of the population at some point in the past

Advantage and **Disadvantage** of sample:

Advantage of sample:

- It takes less time and less cost compared with census.

Disadvantage of sample:

- ❑ Less accurate.
- ❑ May not represent the given population

SAMPLING TECHNIQUE

S

Random Sampling Techniques

Sampling produces significant research result. However, with the differences that can be present between a population and a sample, sample errors can occur. Therefore, it is essential to use the most relevant and useful sampling method.

I

Random Sampling

or

Probability Sampling

Random Sampling Techniques

Random sampling simply describes when every element in a population has an equal chance of being chosen for the sample.

Here's why you should know about **random sampling**.

Random Sampling Techniques

- If you're a data scientist or a researcher and want to develop models, you need data.
- And if you need data, SOMEONE needs collect data.

Random Sampling Techniques

And if someone is collecting data, they need to make sure that it is not biased, or it will be **extremely costly** in the long run.

Therefore, if you want to collect **unbiased data**, then you need to know about random sampling!

Random Sampling Techniques

There are 4 types of random sampling techniques:

- ✓ *Simple Random Sampling*
- ✓ *Systematic Random Sampling*
 - ✓ *Stratified Random Sampling*
- ✓ *Cluster Random Sampling*

Random Sampling Techniques

1. Simple Random Sampling

Simple random sampling requires using randomly generated numbers to choose a sample.

Random Sampling Techniques

2. Systematic Random Sampling

Systematic random sampling is a very common technique in which you sample every k^{th} element.

Random Sampling Techniques

For example, if you were conducting surveys at a mall, you might survey every 100^{th} person that walks in.

Random Sampling Techniques

If you have a sampling frame then you would divide the size of the frame, N , by the desired sample size, n , to get the index number, k .

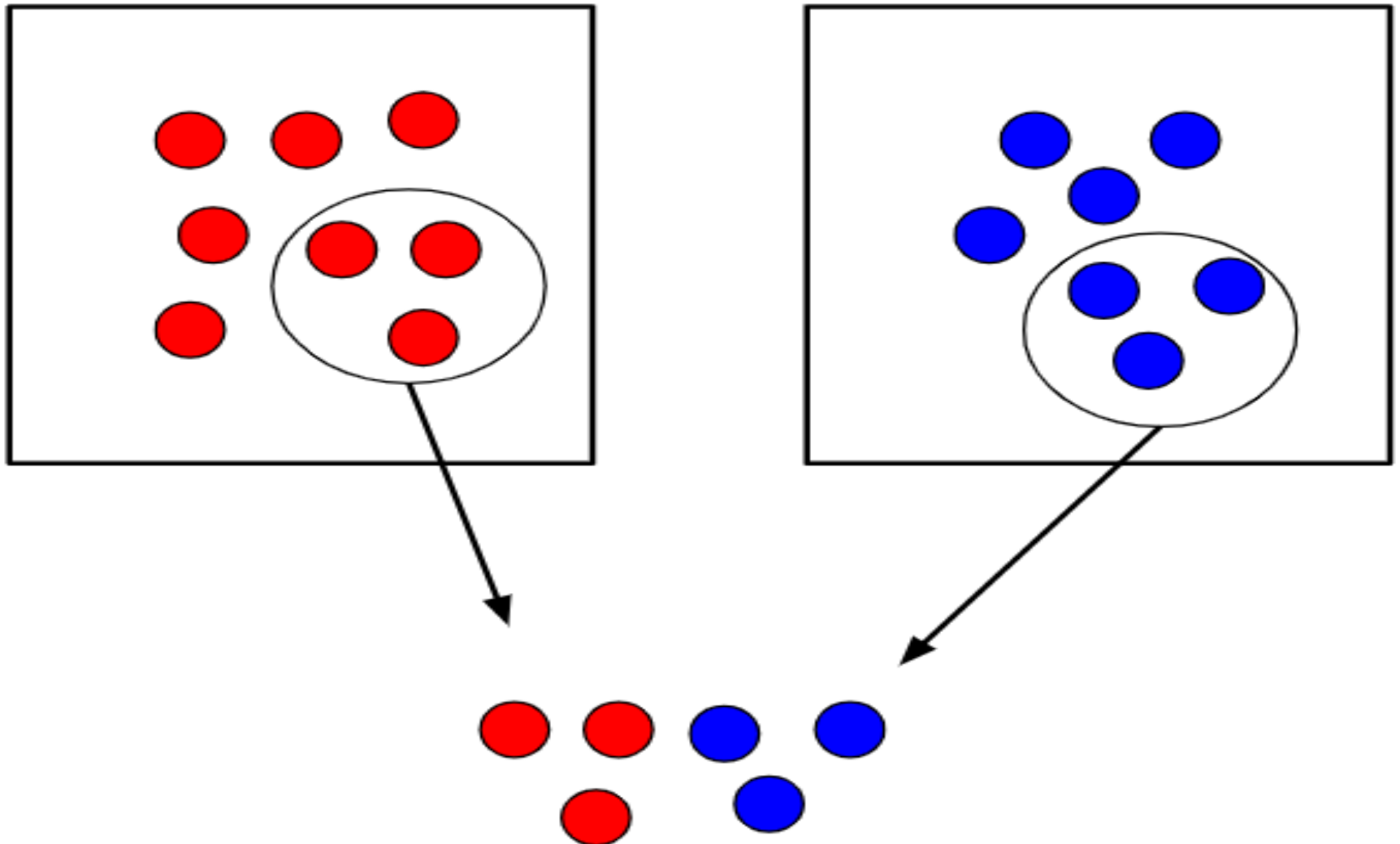
You would then choose every k^{th} element in the frame to create your sample.

Random Sampling Techniques

3. Stratified Random Sampling

Stratified random sampling starts off by dividing a population into groups with similar attributes. Then a random sample is taken from each group.

Random Sampling Techniques



Random Sampling Techniques

This method is used to ensure that different segments in a population are equally represented. To give an example, imagine a survey is conducted at a school to determine overall satisfaction.

Random Sampling Techniques

It might make sense here to use stratified random sampling to equally represent the opinions of students in each department.

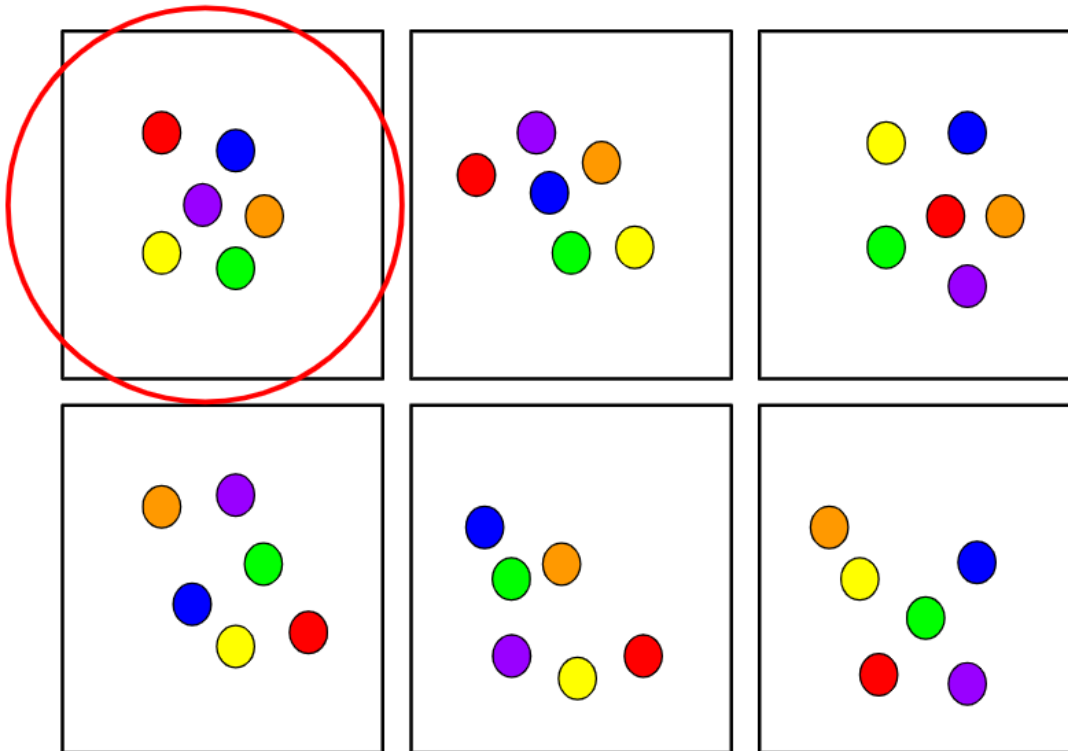
Random Sampling Techniques

4. Cluster Random Sampling

Cluster sampling starts by dividing a population into groups, or **clusters**. What makes this different than stratified sampling is that each cluster must be representative of the population.

Random Sampling Techniques

Then, you randomly selecting entire clusters to sample.

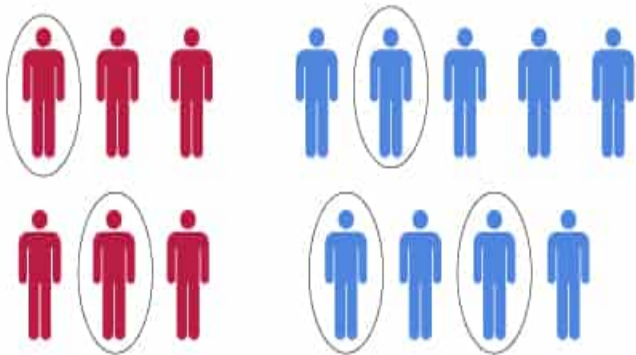


Random Sampling Techniques

For example, if an elementary school had five different grade eight classes, cluster random sampling might be used and only one class would be chosen as a sample.

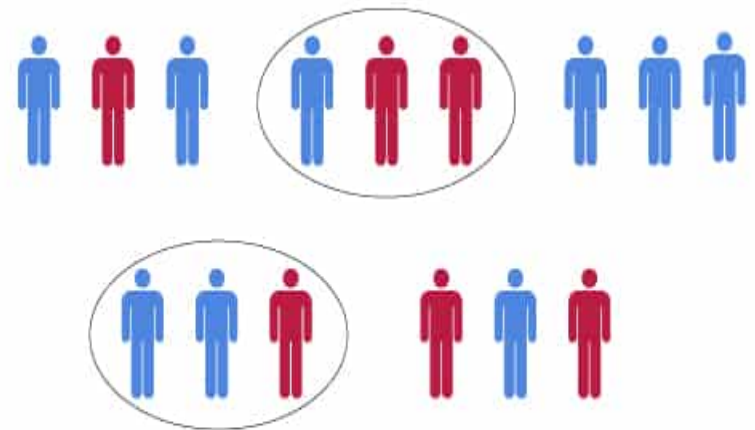
Random Sampling Techniques

Stratified random sampling



VS

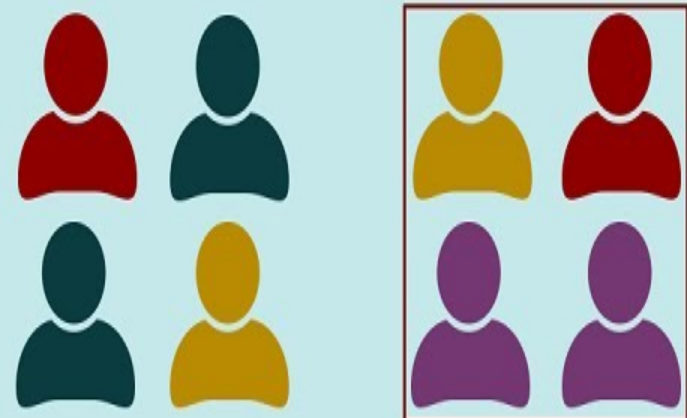
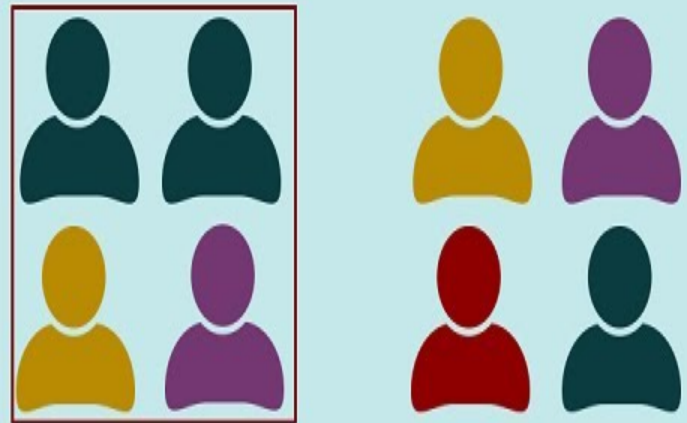
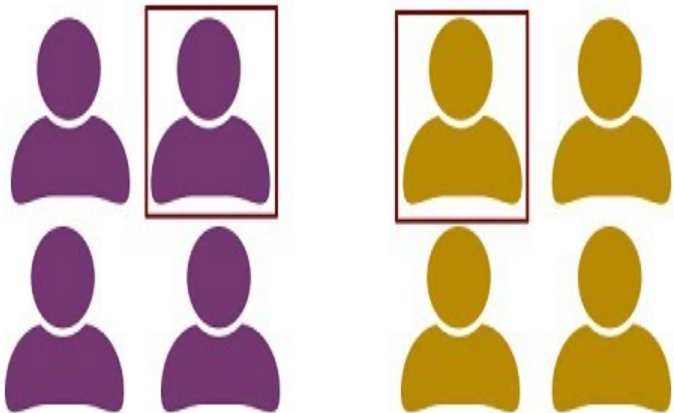
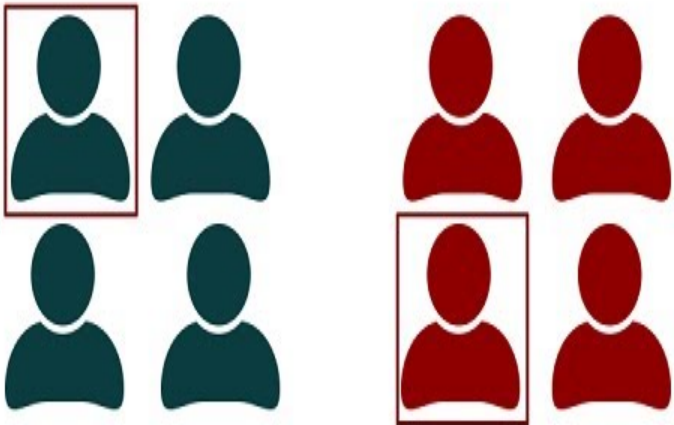
Cluster sampling



Stratified Sampling

Vs

Cluster Sampling



II

Non-Random Sampling

or

Non-Probability

Sampling

Non-Random Sampling Techniques

Non-random or non-probability sampling

is defined as a sampling technique in which the researcher selects samples based on the subjective judgment of the researcher rather than random selection.

Non-Random Sampling Techniques

This sampling method depends heavily on the expertise of the researchers. It is carried out by observation, and researchers use it widely for qualitative research.

Non-Random Sampling Techniques

Non-probability sampling is a sampling method in which not all members of the population have an equal chance of participating in the study, unlike probability sampling. Each member of the population has a known chance of being selected.

Non-Random Sampling Techniques

The following are some of the
non-random sampling methods

✓ *Convenience (or chunk
Sampling)*

✓ *Judgment (or Purposive
Sampling)*

✓ *Quota Sampling*

Non-Random Sampling Techniques

Convenience Sampling

Convenience Sampling also called opportunity or availability sampling is a non-probability sampling technique where samples are selected from the population only because they are *conveniently available* to the researcher.

Non-Random Sampling Techniques

Researchers choose these samples just because they are easy to recruit, and the researcher did not consider selecting a sample that represents the entire population.

Non-Random Sampling Techniques

A classic example of convenience sampling is the *Pepsi Challenge*. Originally, the Pepsi Challenge was a blind taste test conducted at shopping malls, stores, and other public venues. Participants taste unmarked cups containing Coca-Cola and Pepsi and then indicate their preference.

Non-Random Sampling Techniques



Non-Random Sampling Techniques

The Pepsis Challenge has all the hallmarks of this method, including the use of crowded areas facilitated the easy acquisition of participants and the lack of requirements for participating.

Non-Random Sampling Techniques

Judgmental or Purposive sampling

In the Judgmental sampling method, researchers select the samples based purely on the researcher's knowledge and credibility.

Non-Random Sampling Techniques

In other words, researchers choose only those people who they deem fit to participate in the research study.

For example, if you want to study the relationship between cancer rate and nuclear radiation, you might consider for individuals who live near a nuclear power station.

Non-Random Sampling Techniques

Quota Sampling

If the researcher is interested in a particular department within the population the researcher will use quota sampling to divide the population into *strata or groups.*

Non-Random Sampling Techniques

So, quota sampling is the division of the larger population into strata *according to the need of the research.*

Non-Random Sampling Techniques

Snowball sampling

This is a type of convenience sampling in which those participants invited invite other participants and so on to create a pyramid effect.

Non-Random Sampling Techniques



Non-Random Sampling Techniques

It is used where potential participants are hard to find. It's called snowball sampling because (in theory) once you have the ball rolling, it picks up more “snow” along the way and becomes larger and larger.



SOURCE OF
DATA

SOURCE OF DATA

Primary data is a type of information that is obtained directly from first-hand source by surveys, observation or experimentation. It is data that has not been previously published and is derived from a new or original research study.

SOURCE OF DATA

Secondary data is data that is not collected by the person who is doing research. It can be obtained from published and unpublished resources.

SOURCE OF DATA

PRIMARY DATA	SECONDARY DATA
Primary source	Secondary source
Original and new	Re-used and old
Customized as per the need of the research	May not directly linked with research
Less economical	More economical
High on reliability	Low on reliability

SOURCE OF DATA

✓ *Primary Data*

- ✓ Direct Observation
- ✓ Telephone Interview
- ✓ Personal Interview
- ✓ Written Questionnaires

✓ *Secondary Data*

SOURCE OF DATA

i) Direct observation

Observation is way of gathering data by watching behavior, events, or noting physical characteristics in their natural setting.



SOURCE OF DATA

Direct observation is when you watch interactions, processes, or behaviors as they occur; for example, observing a teacher teaching a lesson from a written curriculum to determine whether they are delivering it with fidelity.

SOURCE OF DATA



SOURCE OF DATA

ii) Telephone interview

Telephone interview is a data collection method when the interviewer communicates with the respondent on the telephone in accordance with the prepared questionnaire.

SOURCE OF DATA



SOURCE OF DATA

iii) Personal interview

With respect to primary research, the foremost tool is the personal interview.

The face-to-face contact between researcher and respondent is not equal in terms of the potential quality of data that can be obtained.

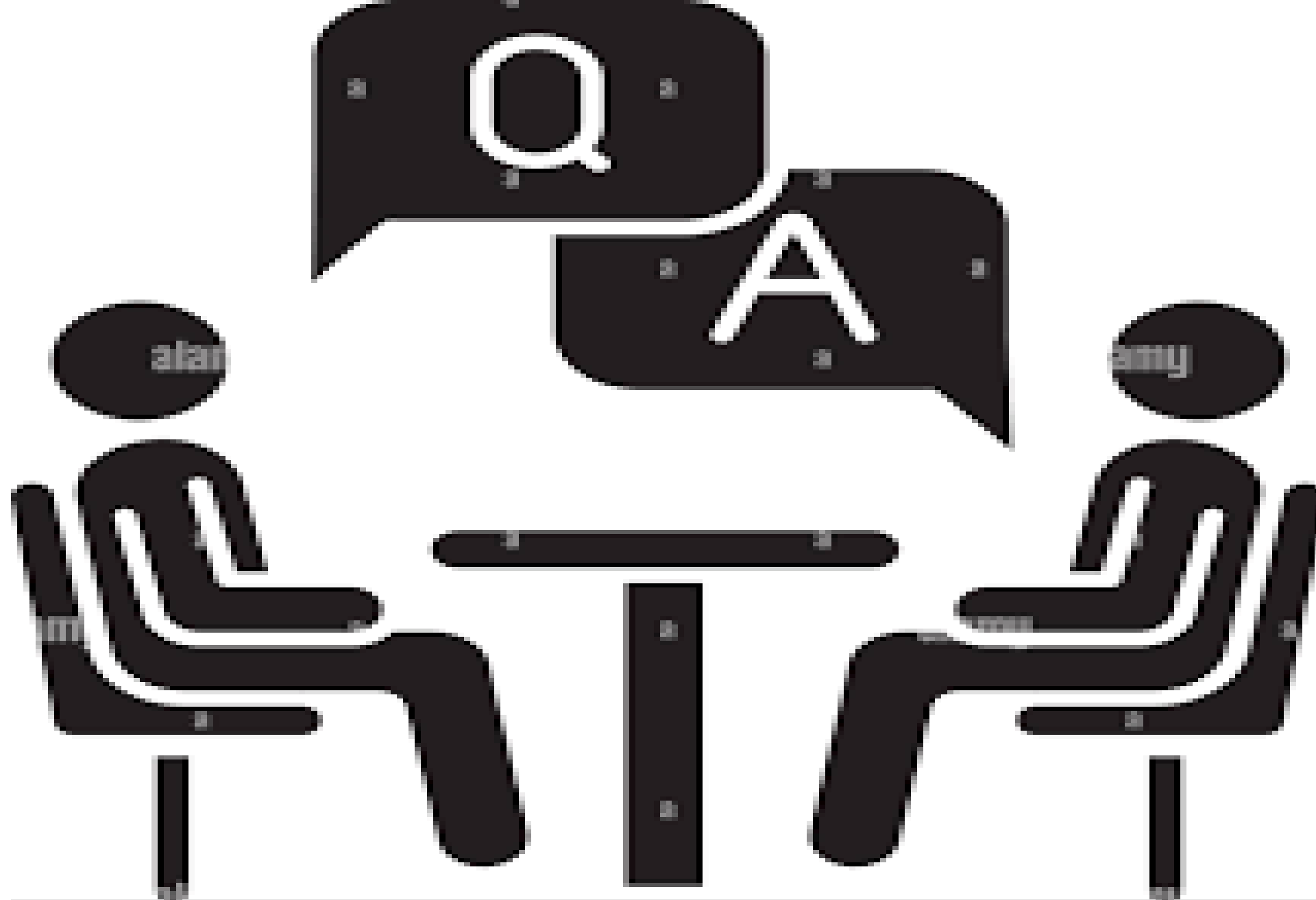


SOURCE OF DATA

In the face-to-face interview it is possible to record more than the verbal responses of the interviewee, which are often superficial. When human beings communicate directly with each other much more information is communicated between them.

SOURCE OF DATA

When two people face one another, dialogue is conducted on several levels. It goes beyond verbal expression. The nature of words used, facial expressions and body language all communicate what the other party means.



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Don't do this in interviewing!



Advisable !



Advisable!



Advisable!



Advisable!



Advisable!



SOURCE OF DATA

iv) Written questionnaires

Written questionnaires are sent directly to the target group members by mail. The questionnaire is then completed by the respondent. This method does not require interviewers; however, it depends highly on the *respondents' motivation to participate* which is reflected in the response rate.

SOURCE OF DATA

Questionnaires are very cost effective when compared to face-to-face interviews.

This is especially true for studies involving large sample sizes and large geographic areas.

SOURCE OF DATA

One major disadvantage of written questionnaires is the possibility of low response rates. Low response is the curse of statistical analysis. It can dramatically lower our confidence in the results.

On a scale of 1 to 5 where 1 is
would you rate your level of

- 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

On a scale of 1 to 5 where 1 is
would you rate your level of

- 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

End of Chapter 2



Any Questions?

